Task 4 — engineering review loop: SPECIFICATION (v7) — ASSIGNMENT FROZEN

Task 4 - engineering review loop - v7 - ASSIGNMENT FROZEN (DRAFT pending named identities)

Status: implemented 2026-09-20. The unlabelled package is built and ready for engineering. No label has been invented and no validation metric has been computed. Every count below was recomputed from the frozen Model 1 and Model 3 artifacts.

v3 applies a second set of corrections of 2026-09-20 — the precision uncertainty statement, the Model 3 validation design, `Unsure` reporting and workbook editability. **v2 applied the first set** — the recall estimand, population reconciliation, the 200-label allocation, and stronger blinding. Each is marked **[1]–[4]** where it lands.

Task 4 is the first point in this project where **ground truth enters**. Everything before it — Model 1's severities, Model 2's WAPE, Model 3's ranking — is unlabelled. How the queue is built therefore determines what can ever be measured.

0. What Task 4 is, and what it is not

Is	A labelling instrument, built so the resulting estimates are unbiased and weightable.
Is	The first ground-truth check either model has had.
Is not	A ranked worklist. A worklist and an estimation sample are different objects.
Is not	Owned by the ML engineer. Engineering owns the judgements; ML owns the instrument.

1. Corrections to the provided Task 4 instructions

#	Provided instruction	Problem	Correction
1	`combined.sort_values(["severity"], ascending=False)`	**Severity is a string.** Descending lexicographic order is `Review Required > Problematic > Critical > Anomaly` — the *least* urgent first, every Critical buried third.	§5
2	`pd.concat([anomaly_review_raw, anomaly_review_triage])`	**Grain mismatch.** Model 1 rows are episodes; Model 3 rows are meter-days.	§3
3	Queue = all of Model 1	**3,928 events** is not a reviewable quantity.	§3, §7
4	"at least 20 labeled events"	20 labels give **±21.9 pp** worst-case, and ~1 label per family × severity cell.	§7
5	"compute precision for Model 1"	A flagged-only queue estimates precision but says nothing about what was missed.	§8
6	"feed confirmed-fault windows into `exclude_windows`"	Would **reintroduce a fixed defect** — masking starved T4's baseline and hid the ZN.16-PB repair. Model 1 is frozen.	§10

2. Population reconciliation [2]

The v1 draft stated **2,834** Model 1 meter-days. **That figure was wrong.** It counted distinct `(meter, event_START date)` pairs, so an episode spanning 500 days contributed a single meter-day. Corrected below.

The observed universe

The observed meter-day universe is a property of `gold.energy_raw` and is identical for both models; Model 3's feature table is simply where it is already materialised.

Population	Full history	Within Model 3's scored universe
Observed meter-days (31 meters)	**15,759**	14,130
Warm-up excluded (Model 3 only)	—	1,629

Model 1 populations

Population	Full history	Within M3 scored
Finding population (meter × phase × detector)	**405**	—
Event population (episodes)	**3,928**	—
Expanded event-days (meter × day × family)	20,056	—
Unique **electrical-fault** meter-days	**9,404**	**8,332**
Unique **non-commissioning** electrical-fault meter-days	**9,404**	**8,332**
Unflagged observed meter-days	**6,356**	5,798

Two things that look like errors and are not:

The two "unique … meter-days" rows are identical by construction. Model 1's severity stage rewrites `event_class` to `COMMISSIONING` for any commissioning episode, so no `ELECTRICAL_FAULT` row carries `commissioning = True`. The two rows measure the same set under that encoding; they are shown separately only because the correction asked for both.

Flagged meter-days are not a strict subset of observed meter-days. 9,404 flagged but only **9,403** are observed: ZN.19-PB.1 on 2025-10-31 lies inside a Model 1 episode whose start and end bracket a day on which that meter reported nothing (its next observed day, 2025-11-01, has 11 readings). Hence `9,403 + 6,356 = 15,759` exactly.

> **The sampling frame is therefore `observed ∩ flagged` = 9,403 and `observed − flagged` = 6,356.** A day with no telemetry cannot be shown to a reviewer.

By family (non-commissioning electrical-fault meter-days):

Family	Full history	Within M3 scored
T1 polarity	3,098	2,771
T2 current/power	1,265	1,106
T3 power factor	7,693	6,896
T5 register	605	474

Excluded from "flagged" and carried only as context: `COMMISSIONING` 88 meter-days, `DATA_AVAILABILITY` 2,119, `ELECTRICAL_CHANGE` 1,420.

Which universe the control stratum draws from [2]

The estimand concerns **Model 1**, so the frame is **Model 1's own applicable history**: the 15,759 observed meter-days, giving **6,356 unflagged**. Model 3's warm-up-restricted 14,130 is *not* used, because Model 3's warm-up is a property of Model 3's training window and has nothing to do with what Model 1 did or did not flag. The 5,798 figure is recorded only so the two are never confused.

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3. The review unit, and four strata

Two review units, never blended:

- **Finding-level** for Model 1 — "*is ZN.12-PB phase 3's reverse power a real fault?*" is one judgement, not 18 about 18 episodes. 405 findings exist; **255 assert a fault** and form the stratum A frame (§7.1).
- **Meter-day-level** for Model 3 and the control — "*was this meter's behaviour on this day genuinely unusual?*"

Stratum	Grain	Source	Population	Estimand
A	finding	Model 1 **ELECTRICAL_FAULT** findings, T1/T2/T3/T5 (§7.1)	**255**	Model 1 **precision**, by family and severity
B	meter-day	Model 3 Top-100 high-ranked **plus same-meter reference days** (§7A)	99 high / 671 refs	whether Model 3's ranking tracks confirmed faults, **within meter**
C	meter-day	Model 3 List C, unexplained	**1**	whether Model 3 surfaces anything Model 1 misses
D	meter-day	Model 1 **unflagged** observed meter-days	**6,356**	`unflagged_fault_rate` — §4

Stratum C is kept although it holds a single item; its size *is* the finding — at Top-100, Model 1 explains 99 of Model 3's 100 highest-ranked meter-days.

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4. The recall estimand, corrected [1]

The v1 draft called stratum D a measure of "false negatives / recall". **That was wrong.**

Stratum D estimates one quantity and only one:

```
unflagged_fault_rate  =  P( engineering confirms a fault | Model 1 did not flag that meter-day )
```

This is a **false omission rate**. It is *not* a false-negative rate and *not* Model 1 recall. The difference is not terminological:

Quantity	Denominator	Obtainable from D?
`unflagged_fault_rate` (false omission rate)	**unflagged** meter-days	**yes**
False-negative rate	**all true faults**	no

Quantity	Denominator	Obtainable from D?
Recall = TP / (TP + FN)	**all true faults**	no

Recall needs the count of true faults, which requires labelling flagged **and** unflagged units *at the same grain*. Stratum A is finding-level and stratum D is meter-day-level, so:

> **Strata A and D must never be combined into a recall calculation.** A precision computed over the **255 ELECTRICAL_FAULT finding frame** (§7.1) and an omission rate computed over 6,356 meter-days have different denominators, different units and no common divisor. Dividing one by the other produces a number with no meaning.

If meter-day recall is wanted later it requires a separately designed, same-grain sample drawing from *both* the 9,403 flagged and the 6,356 unflagged observed meter-days, with known inclusion probabilities in each. That is a different instrument and is explicitly out of scope for Task 4 v1.

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5. Ordering, and the severity bug

```
SEVERITY_RANK = {"Critical": 0, "Problematic": 1, "Anomaly": 2, "Review Required": 3}
worklist.sort_values("severity", key=lambda s: s.map(SEVERITY_RANK))
```

The provided `sort_values(["severity"], ascending=False)` yields `Review Required → Problematic → Critical → Anomaly`.

The estimation samples are not sorted by severity at all — they are randomised (§6). Ordering a sample by the thing it measures invites reviewers to stop early and truncates the tail the estimate depends on. Severity ordering belongs to the operational worklist (§9).

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6. Blinding [4]

Two **separate randomised sheets**, because the two grains need different blinding regimes.

Sheet 1 — finding-level estimation sample (stratum A)

Every row here is flagged by definition; there is no flag status to conceal. What is concealed is the model's **verdict**.

Shown	Withheld to the key
meter, phase, date range	severity
measured electrical evidence	magnitude band, corroboration verdict
concurrent production classifier status	Model 1's own confidence ordering

> **Stated limitation.** At finding level the detector family cannot be fully blinded, because the evidence a reviewer needs — negative kW with positive current, or a frozen register under load — identifies the family by itself. What is blinded is severity and the model's confidence. Claiming full blinding here would be false.

Sheet 2 — meter-day estimation sample (strata B + C + D)

Flagged and unflagged rows are mixed and **must expose exactly the same visible schema**, or the control is worthless.

Withheld without exception:

- Model 1 flag status
- detector family
- severity
- Model 3 score and rank
- stratum
- **any model-written description**, because a sentence like "*reverse power from 2026-06-03*" reveals the row was flagged

Shown: a neutral row identifier, meter, Saudi date, and raw electrical evidence computed **uniformly for every row from Gold** — reading count, coverage, mean load %CB, per-phase current and power summaries, power-factor summaries, negative-power counts, production classifier status counts.

> **Design consequence.** Because the evidence must be computed the same way for flagged and unflagged rows, it **cannot** be lifted from Model 1 event records — those exist only for flagged rows and their presence would be the tell. Sheet 2's evidence columns are derived from the Model 3 feature table and raw Gold aggregates, which exist for every observed meter-day.

The key (`task4_queue_key.parquet`) is held by the ML side and joined back only after labels return.

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7. Allocation, and the frozen stratum A design

7.1 Stratum A frame — a correction

Stratum A asks "*is this a confirmed fault?*". That question only applies to findings that **assert** a fault, and not every Model 1 finding does:

Family	ELECTRICAL_FAULT	COMMISSIONING	ELECTRICAL_CHANGE	DATA_AVAILABILITY
T1	14	16	0	0
T2	72	12	0	0
T3	137	0	0	0
T5	32	3	0	0
T4	**0**	0	**88**	**31**

T4 asserts no fault at all — its 119 findings are level shifts and telemetry outages. Asking engineering to confirm a telemetry outage as a fault is a category error, so T4 is excluded from stratum A and routed to the operational worklist instead.

The 31 commissioning findings are also held out: Model 1 explicitly does not assert them as faults and caps them at Review Required. Testing the 2024-12-31 commissioning assumption is worthwhile but is a **different estimand** needing its own sub-sample — recorded as an open item (§14) rather than silently folded into a precision figure.

> **Stratum A frame = 255 ELECTRICAL_FAULT findings across T1, T2, T3, T5.**

7.2 The frozen allocation algorithm

Fixed before any sampling, and fully deterministic:

1. `base = floor(110 / 4) = 27`. Give each family `min(base, N_family)`.
2. Redistribute any shortfall one unit at a time to the family with the most unused room, ties broken by family name.
3. Within a family, give every non-empty severity cell `min(2, N_h)`, then award the remainder one unit at a time to the cell with the most unused room, ties broken by cell size then severity name.
4. `pi_h = n_h / N_h` is the inclusion probability for **every finding** in cell *h*.

Frozen allocation and inclusion probabilities:

Family	Severity	N_h	n_h	pi_h	weight 1/pi_h
T1	Anomaly	1	1	1.0000	1.00
T1	Critical	12	12	1.0000	1.00
T1	Review Required	1	1	1.0000	1.00
T2	Anomaly	24	7	0.2917	3.43
T2	Critical	2	2	1.0000	1.00
T2	Problematic	13	2	0.1538	**6.50**
T2	Review Required	33	16	0.4848	2.06
T3	Anomaly	40	11	0.2750	3.64
T3	Critical	38	9	0.2368	4.22
T3	Problematic	48	20	0.4167	2.40
T3	Review Required	11	2	0.1818	5.50
T5	Critical	32	27	0.8438	1.19
	total	**255**	**110**		

Family totals: **T1 14 of 14 (a census)**, T2 27 of 72, T3 42 of 137, T5 27 of 32. Inclusion probabilities span **0.1538 to 1.0000**, so design weights span **1.00 to 6.50**.

7.3 The precision interval — corrected

The v2 draft claimed +/-6.9 pp from 200 labels. That was wrong. The 200-label budget is the whole project; **only stratum A's 110 labels estimate Model 1 precision.**

Statement	Status
+/-6.9 pp from 200 labels	**withdrawn**
+/-9.3 pp from 110 simple-random observations	**intuition only, not the reported interval**
Survey-weighted linearisation with FPC	**the reported interval**
FPC-aware stratified bootstrap	secondary cross-check only

Stratum A is stratified with **unequal inclusion probabilities spanning 6.5x**, so a simple binomial interval is the wrong variance. The point estimate is a weighted ratio estimator over resolved cases,

```
precision_hat = sum(w_i * y_i) / sum(w_i),    w_i = 1 / pi_h(i)
```

The reported interval is survey-weighted linearisation with a finite-population correction. For a ratio estimator under stratified sampling without replacement,

```
Var(R) ~= (1 / (sum w)^2) * sum_h [ N_h^2 * (1 - f_h) / n_h * s2_h ],   f_h = n_h / N_h
```

with `e_i = y_i - R` and `s2_h` the within-cell sample variance of `e`. The `(1 - f_h)` factor is **exactly zero when** `f_h = 1`, so a census cell contributes **no sampling variance at all** — which is correct, because every unit in it was examined. Four of the twelve cells are censuses (all three T1 cells and T2 Critical).

A **finite-population-aware stratified bootstrap** runs as a **secondary cross-check**: it holds census cells **fixed** across replicates rather than resampling them, and rescales each non-census cell's deviations by `sqrt(1 - f_h)`. It is reported alongside the primary interval, never in place of it.

A simple binomial interval is never the reported figure. Quoting `1.96*sqrt(p(1-p)/n)` would treat a stratified design with inclusion probabilities spanning 6.5x as though it were a simple random sample, and would additionally attribute sampling variance to cells that have none.

The +/-9.3 pp figure appears once, labelled "*simple-random intuition only*", and is never used as the interval.

7.4 Full allocation

Stratum	Rule	Today	Population	Limitation
A	fixed 110, per §7.2	110	255	none; T1 is a census at 14 of 14
D	fixed 40	40	6,356	none
C	census, capped at 20	**1**	1	**population-limited** — one unexplained item at Top-100
B	**50 - C**, split high-ranked / reference	**49** = 25 + 24	99 high, 671 refs	none
	total	**200**		

Any stratum unable to fill its allocation records `allocation_requested`, `allocation_available` and `allocation_drawn`, and is **never back-filled** from another stratum — that would break the inclusion probabilities.

7.5 What the other strata buy

For stratum D, 40 controls with zero confirmed faults put the 95% upper bound on `unflagged_fault_rate` at **7.2%**; at 20 controls it would be only 13.9%.

Per-family precision at 27 labels is directional only and is always reported with its linearised FPC interval.

7A. Stratum B — the Model 3 validation design

Stratum D is not the Model 3 control. D is a uniform random draw from the 6,356 Model-1-unflagged meter-days; its estimand is `unflagged_fault_rate`, and it is spread thinly across all 31 meters, so it cannot answer "did Model 3 rank the right days *within a meter*".

The Model 3 budget stays at **50**: `C` = census of unexplained Top-100 items (capped at 20), and the remaining `50 − C` is a **blinded mixture of high-ranked days and same-meter lower-ranked reference days**. Today C = 1, giving **25 high-ranked + 24 reference**.

The reference-selection rule, frozen before any label is seen

A reference day must:

1. come from the **same meter** as a selected high-ranked day;
2. lie in that meter's **own bottom quartile** of `window_anomaly_pct`;
3. be **at least 14 days** from every high-ranked day selected for that meter, so a reference cannot fall inside the same episode;
4. come from the 14,130-day scored universe.

Why the contrast is relative, not absolute. The v2 sketch implied an absolute gap ("sufficiently lower"). Measured, an absolute 0.40 gap is **impossible** for two of the six meters involved:

Meter	min `window_anomaly_pct`	q25	refs under an absolute 0.40 gap	refs under the bottom-quartile rule
ZN.10-PB.1	**0.707**	0.855	**0**	117
ZN.11-PB.2	0.370	0.809	3	106
ZN.12-PB	0.044	0.646	50	113
ZN.10-PB.2	0.009	0.422	125	96
ZN.1-PB.1	0.003	0.560	123	123
ZN.16-PB	0.306	0.907	14	116

ZN.10-PB.1 has never scored below 0.707 in its entire history, so no day of that meter is 0.40 below its high-ranked days. An absolute floor would silently drop the meter from the validation altogether. The bottom-quartile rule yields **671 available references against the 24 needed**.

The realised contrast is recorded per pair, not assumed. Minimum realised contrast by meter: ZN.10-PB.2 0.577, ZN.1-PB.1 0.440, ZN.12-PB 0.353, ZN.11-PB.2 0.190, ZN.10-PB.1 0.144, ZN.16-PB 0.091. The ZN.16-PB pair is thin and is reported as such rather than dropped, since that meter contributes a single high-ranked item.

Analysis — within meter, never pooled

For each meter, compare confirmed-fault outcomes for its high-ranked days against its own reference days, then combine across meters with a **stratified (per-meter) procedure** — an exact conditional test such as Mantel-Haenszel, or a per-meter paired bootstrap.

A pooled rank-versus-label statistic is **forbidden**. The corrected permutation null already demonstrated that pooling recovers meter composition rather than day selection: T1 and T3 significance reversed entirely once composition was held fixed, and only T2 survived.

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8. What labels can and cannot establish

Quantity	Estimable?	From
Model 1 precision, overall	**yes**	A, weighted ratio estimator + survey-weighted linearisation with FPC (§7.3)
Model 1 precision, per family	directional only, with its linearised FPC interval	A
`unflagged_fault_rate` (false omission rate)	**yes**	**D alone**
Model 1 **recall**	**no** — needs a same-grain flagged+unflagged design (§4)	—
Model 3 ranking validity	**yes, within meter**	**B only** — high-ranked vs same-meter reference days (§7A). **Not D.**
Model 2 forecast accuracy	**no** — forecasting is not fault detection	—

The Model 3 row carries a constraint from the corrected permutation null: Model 3's apparent agreement with Model 1 was **largely meter composition, not day selection** — T1 and T3 significance reversed once composition was held fixed, and only T2 survived. Any label-based validation of Model 3 must therefore be computed **within meter**, or it recovers the same artifact.

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9. Two outputs, not one

Artifact	Purpose	Ordering	Blinded
`task4_labelling_sample.xlsx` — sheet 1 (findings), sheet 2 (meter-days)	estimation	randomised within each sheet	**yes**, per §6
`task4_worklist.xlsx`	fixing things	severity rank, then Model 3 rank	**no**

`.xlsx` for both: engineering edits directly and it avoids CSV encoding round-trips.

Editability, resolved. One workbook is generated **per reviewer**, with `reviewer_id` **pre-populated and locked** — if reviewers could type it, the 10% double-labelled overlap sample could be mis-attributed and the kappa estimate would be meaningless. The file name and the key both carry the same reviewer id, so attribution is verifiable without trusting a cell.

Column	Locked?
`review_id`, all context and evidence columns	locked
`reviewer_id`	**locked, pre-populated**
`confirmed_fault`, `fault_type`, `notes`, `reviewed_at`	**unlocked**

`reviewed_at` is unlocked because only the reviewer knows when they judged a row, and a locked cell cannot be auto-stamped without macros. If it is returned blank, ingest falls back to the workbook's return timestamp and sets `reviewed_at_imputed = True` rather than inventing a date silently.

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10. The feedback loop, corrected

Feeding confirmed faults into `model1_config.yaml`'s `exclude_windows` **must not be implemented**. Model 1 v1 is frozen, and masking known fault windows is precisely what starved T4's baseline and made the ZN.16-PB repair undetectable; the frozen implementation deliberately uses an unmasked trailing baseline with `baseline_gap_days: 7`.

Labels instead feed: the **acceptance set** (a confirmed Tier-B finding is promoted to Tier A with an engineering reference); the **commissioning boundary** (2024-12-31 is currently an assumption); the **severity definition** (whether confirmed faults cluster by band); and a **future Model 1 v2**, which would need its own freeze.

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11. Label schema

Column	Filled by	Values
`review_id`	ML	stable opaque id; the only join key
neutral context and evidence columns	ML	§6
`confirmed_fault`	**engineering**	`Y` / `N` / `Unsure`
`fault_type`	**engineering**	free text or short controlled list
`notes`	**engineering**	free text
`reviewer_id`	**pre-populated per copy, locked**	see below
`reviewed_at`	**engineering, unlocked**	date

`Unsure` is a first-class outcome and is **never silently discarded**. Four quantities are reported together, always:

Reported quantity	Definition
resolved-case estimate	weighted precision over `Y` and `N` only, `Unsure` excluded from the denominator
weighted `Unsure` rate	design-weighted share of sampled findings labelled `Unsure`
conservative bound	every `Unsure` counted as `N`
optimistic bound	every `Unsure` counted as `Y`

The conservative and optimistic bounds bracket the truth whatever the unresolved cases turn out to be, so a high `Unsure` rate widens the reported range rather than quietly improving the headline. Reporting only the resolved-case figure would inflate precision by removing exactly the hardest cases.

A **10% overlap sample** is double-labelled to estimate Cohen's kappa; a precision figure is only as good as the reproducibility of the judgement behind it.

A named engineering adjudicator is required before collection begins (approved). They resolve `Unsure` and inter-reviewer disagreement.

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12. Tests

Area	Criterion
Severity ordering	`Critical` sorts first; a string sort is asserted **wrong**, so the bug cannot return
Estimand separation	no artifact or function combines stratum A with stratum D into a recall figure
Naming	the string `recall` never appears on a stratum-D output; the field is `unflagged_fault_rate`
Control frame	stratum D draws from the **6,356** full-history unflagged meter-days, not the 5,798 scored subset
Frame integrity	every sampled meter-day is an **observed** meter-day; the ZN.19-PB.1 2025-10-31 case is excluded
Reconciliation	9,403 observed-flagged + 6,356 unflagged = 15,759, asserted
Allocation	requested total is exactly 200 for any C in 0..20; `B = 50 − C`
Shortfall	a stratum unable to fill its allocation records it and is never back-filled
Blinding, sheet 2	contains no flag status, family, severity, Model 3 score/rank, stratum or model-written description
Schema identity	flagged and unflagged rows in sheet 2 have identical column sets and no tell-tale nulls
Sheet separation	finding-level and meter-day-level rows never share a sheet
Inclusion probabilities	every sampled row carries a non-zero recorded probability; weights sum to the stratum population
Join integrity	every returned label re-joins to exactly one key row; unmatched labels fail loudly
`Unsure`	all four quantities reported: resolved-case, weighted rate, conservative (all N), optimistic (all Y)
`Unsure`	no code path drops an `Unsure` row without recording it
Stratum A frame	contains only `ELECTRICAL_FAULT` findings in T1/T2/T3/T5; no T4 and no commissioning finding
Allocation determinism	the frozen algorithm reproduces the §7.2 table exactly from the findings file
Inclusion probabilities	every sampled finding carries `pi_h = n_h / N_h`; weights span 1.00 to 6.50
Precision interval	the reported interval comes from **survey-weighted linearisation with FPC**, never from `1.96*sqrt(p(1−p)/n)`
Precision interval	a census cell (`pi_h = 1`) contributes **exactly zero** sampling variance
Precision interval	the FPC-aware bootstrap is a **secondary cross-check** and holds census cells fixed
Precision interval	the simple +/-9.3 pp figure is labelled "simple-random intuition only" wherever it appears
Stratum B references	every reference is the **same meter**, in that meter's own bottom quartile, and >= 14 days from every high-ranked day for that meter
Stratum B contrast	the realised contrast is recorded per pair, and no absolute gap floor is imposed
Stratum B analysis	no pooled rank-vs-label statistic is computed; the test is stratified by meter

Area	Criterion
Stratum D isolation	stratum D rows are never used in the Model 3 ranking validation
Workbook editability	`reviewer_id` is pre-populated and locked; `confirmed_fault`, `fault_type`, `notes` and `reviewed_at` are unlocked; every other cell is locked
Workbook editability	a returned blank `reviewed_at` sets `reviewed_at_imputed = True` rather than inventing a date
Idempotence	rebuilding from the same artifacts and seed is byte-identical
No write-back	Task 4 writes nothing into `model1_config.yaml` or any frozen artifact

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13. Definition of Done

13.1 Built, tested and frozen

- [x] `src/review_loop/build_queue.py` — strata, recorded inclusion probabilities, blinding, both workbooks
- [x] `config/task4_config.yaml` — budget, the frozen allocation rule, seed, quotas, reference-selection rule, assignment contract
- [x] `outputs/<ts>/task4_allocation_stratum_a.csv` and `task4_allocation_stratum_b.csv` — the frozen family x severity table with `N_h`, `n_h` and `pi_h`, written **before** any row is drawn, alongside `task4_allocation_report.csv` (requested / available / drawn)
- [x] `outputs/<ts>/task4_worklist.xlsx` — un-blinded, severity-ordered
- [x] `outputs/<ts>/task4_queue_key.parquet` — withheld key, updated with assignments
- [x] `src/review_loop/assignment.py` — reviewer assignment frozen: 200 unique units, 20 double-reviewed, 110 / 110 workload, reproducible from seed `20260926`
- [x] `tests/test_task4.py`, `tests/test_task4_assignment.py` — §12 passing
- [x] `run_metadata.json` — population reconciliation, allocation requested/available/drawn per stratum, seed, and explicit statements that no label exists yet, that no precision or recall is claimed, that stratum D measures `unflagged_fault_rate` only, and that the precision interval will come from a survey-weighted linearisation with finite-population correction, cross-checked by an FPC-aware stratified bootstrap, and never from a simple binomial

13.2 Blocked on the named reviewer and adjudicator identities

These stay unchecked **by design**, not because work is outstanding on my side. Each one needs the two real reviewer identifiers and the named adjudicator; none of them requires a model rerun, a redraw, or a change of assignment.

- [ ] `config/task4_config.yaml` — real reviewer identifiers replacing `reviewer_1` and `reviewer_2`, and the named adjudicator replacing `adjudicator: TBD`
- [ ] `run_metadata.json` — named adjudicator recorded
- [ ] `outputs/<ts>/task4_labelling_sample_<reviewer>.xlsx` — one per reviewer, two randomised sheets, `reviewer_id` pre-populated and locked. **Built and verified, but written only as `_DRAFT` copies under placeholder identifiers**: `issuable` is `false` and `issue_workbooks.py` exits non-zero until the identities are supplied
- [ ] Delivered to engineering; **collection is theirs**

Estimation from returned labels is a **separate stage**.

14. Decisions

#	Decision	Status
1	Target 200 labels	**approved**
2	Blinding	**approved**
3	Blind control stratum	**approved**
4	Model 1 review unit = finding	**approved**
5	Labels cannot modify frozen Model 1 v1	**approved**
6	Named engineering adjudicator required before collection	**approved**
7	Stratum D estimand is `unflagged_fault_rate`, never recall	**corrected in v2**
8	Control frame is Model 1's own 15,759-day universe	**corrected in v2**
9	`A=110, D=40, C=census≤20, B=50−C`	**corrected in v2**
10	Sheet-2 blinding withholds family, description and flag status	**corrected in v2**
11	Precision interval: +/-9.3 pp is intuition only; +/-6.9 pp withdrawn	**corrected in v3**
11a	Primary CI is **survey-weighted linearisation with FPC**; the FPC-aware stratified bootstrap is a **secondary cross-check**; a simple binomial CI is never reported	**corrected in v7**
12	Stratum A frame = 255 ELECTRICAL_FAULT findings; T4 and commissioning findings excluded	**corrected in v3**
13	Family x severity allocation and every inclusion probability frozen before sampling	**corrected in v3**
14	Stratum B = 25 high-ranked + 24 same-meter bottom-quartile references; D is not the Model 3 control	**corrected in v3**
15	Reference contrast is within-meter relative, not an absolute gap	**corrected in v3**
16	`Unsure` reported four ways, never discarded	**corrected in v3**
17	`reviewer_id` pre-populated and locked per copy; `reviewed_at` unlocked	**corrected in v3**
18	Commissioning-assumption check needs its own sub-sample	**open**

15. Final contract corrections, v4

15.1 Abnormality is not the same thing as a fault

Model 3 is an **anomaly-triage ranker**, not a fault classifier. Asking "was this a confirmed fault?" about a Model 3 item measures the wrong thing: a day can be genuinely abnormal — a shutdown, a commissioning visit, a telemetry outage — without being an electrical fault, and Model 3 is not claiming otherwise.

The meter-day sheet therefore carries **three** label columns:

Column	Values
`engineering_abnormality`	`Y` / `N` / `Unsure`
`electrical_fault`	`Y` / `N` / `Unsure`
`abnormality_type`	Electrical fault / Data availability / Operational change / Commissioning / Normal / Other

Estimand	Outcome used
Stratum D `unflagged_fault_rate`	`electrical_fault`
Stratum B within-meter high vs reference, **primary**	`engineering_abnormality`
Stratum B electrical-fault enrichment, **secondary**	`electrical_fault`

The four-way `Unsure` reporting (resolved-case, weighted rate, conservative, optimistic) is applied **separately to whichever outcome is being estimated**, never pooled across the two.

The findings sheet keeps `confirmed_fault`, because Model 1 *does* assert a fault and that is the claim under test.

15.2 The allocation is integer-exact

`base = floor(110 / 4) = 27`, then the documented one-unit-at-a-time redistribution. It reproduces the frozen table exactly and is asserted on every run:

T1 = 14, T2 = 27, T3 = 42, T5 = 27, total 110, over the 255-finding frame, with inclusion probabilities from **0.1538 to 1.0000**. The build refuses to continue if either the family totals or the total differ from the frozen values.

15.3 The precision interval is finite-population aware

A naive with-replacement bootstrap is **wrong for this design**. Four of the twelve cells are censuses — all three T1 cells and T2 Critical — and a census has **zero sampling variance**, because every unit was examined. Resampling it would invent variability that does not exist and inflate the interval.

Property	Implementation
Point estimate	weighted ratio `sum(w·y) / sum(w)`, unchanged
Primary interval	survey-weighted linearisation with FPC: the `(1 − f_h)` factor is exactly 0 when `f_h = 1`
Cross-check	stratified bootstrap that holds census cells **fixed** across replicates and rescales non-census cells by `sqrt(1 − f_h)`
Singleton non-census cells	reported by name, never silently dropped
`Unsure` bounds	unchanged, applied per outcome

Tested: a census-only design returns `se = 0.0` and a bootstrap interval of zero width; a high sampling fraction yields a strictly narrower interval than a low one.

15.4 Stratum B sampling, frozen before any row was drawn

Decision	Rule	Result
High rows from the 99	eligible meters only (≥2 high pool **and** ≥2 reference pool), floor 2, cap 6, remainder by most unused room	**25 across 7 meters**
High rows per meter	ZN.10-PB.1 4, ZN.10-PB.2 4, ZN.11-PB.2 4, ZN.12-PB 4, ZN.16-PB 3, ZN.17-PB 4, ZN.8-PB.2 2	π from 0.1667 to 1.0000
Reference rows	floor 1 per participating meter, remainder by most unused room, capped at 3× that meter's high allocation	**24**
Reference pool	same meter, that meter's own bottom quartile, ≥14 days from every high-ranked day of that meter	π from 0.0081 to 0.0746
Seeds	`stratum_b_high: 20260922`, `stratum_b_reference: 20260923`	fixed in config

Selection is **random within each frozen cell**: the allocation decides how many, the seed decides which. Three meters (ZN.1-PB.1, ZN.5-PB, ZN.7-PB) contribute a single high-ranked row each and are excluded, because one row cannot support a within-meter comparison.

No selection decision anywhere depends on an engineering label, and none could — the label columns are empty.

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16. Build record, 2026-09-20

Stratum	Requested	Available	Drawn	Shortfall
A — Model 1 findings	110	255	**110**	0
B — Model 3 high + reference	49	1,121	**49**	0
C — unexplained census	20	**1**	**1**	19 (population-limited)
D — blind control	40	6,356	**40**	0
			200	

Population reconciliation, asserted at build time: 15,759 observed meter-days = 9,403 observed-and-flagged + 6,356 unflagged, with one flagged day (ZN.19-PB.1, 2025-10-31) falling on a day that meter reported nothing.

Artifacts — `outputs/task4_20260920T102354Z/`:

File	Contents
`task4_allocation_stratum_a.csv` / `_b.csv`	the frozen allocations, written **before** sampling
`task4_allocation_report.csv`	requested / available / drawn / shortfall per stratum
`task4_labelling_sample_reviewer_1.xlsx`, `_reviewer_2.xlsx`	blinded, two sheets, protected
`task4_worklist.xlsx`	un-blinded, all 405 findings, severity-ordered
`task4_queue_key.parquet`	withheld key, 200 rows

File	Contents
`run_metadata.json`	reconciliation, allocations, digests of every frozen artifact

Sheet shapes: findings 110 rows × 13 visible + 4 label columns; meter_days 90 rows × 21 visible + 5 label columns. All label cells are empty and unlocked; every other cell is locked; `reviewer_id` is pre-populated and locked.

Still blocking collection: the adjudicator is recorded as `TBD` and must be named before the workbooks are issued.

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17. Reviewer assignment contract

The 200 sampled units, their inclusion probabilities and the allocation tables are **frozen and untouched**. Assignment decides *who* reviews a unit; it cannot change which units exist or what any of them weighs. The build verifies that explicitly and refuses to re-issue if a single inclusion probability moves.

17.1 The rule

Assignment is stratified by **(sheet, stratum)**, so neither reviewer receives a skewed mix and the overlap sample spans strata rather than landing entirely in one:

1. Allocate the ~10% overlap across (sheet, stratum) in proportion to size, by largest remainder, ties broken on stratum size then position.
2. Within each group, permute with the fixed seed; the first `n_overlap` become double-reviewed.
3. Alternate the remainder between the two reviewers, flipping which reviewer starts on each group so workloads stay level.

Seed `20260926`, fixed in `config/task4_config.yaml`. Re-running reproduces the assignment exactly; a different seed produces a different one. Both are tested.

17.2 What was assigned

	Units	Sole	Double-reviewed
findings — reviewer 1	61	50	11
findings — reviewer 2	60	49	11
findings, unique	**110**	99	**11**
meter_days — reviewer 1	49	40	9
meter_days — reviewer 2	50	41	9
meter_days, unique	**90**	81	**9**
total unique	**200**	**180**	**20**

20 of 200 double-reviewed — exactly the 10% target. Total assignments 220. Reviewer 1 holds 110 units, reviewer 2 holds 110: neither receives all 200. **The union of reviewer assignments covers all 200 unique units; 20 units are intentionally assigned to both reviewers.**

Overlap by stratum: findings **A 11**, meter-days **B 4**, **C 1**, **D 4**.

17.2a Stratum C is forced into the overlap

Proportional allocation gave the single stratum C unit no overlap. Because it is the **only unexplained Model-3 Top-100 item**, it is now forced into the double review as a **governance and quality check**, and the total is held at exactly 20 by releasing one stratum B overlap.

Which B unit is released follows from the seed already in use, not a new decision: the per-group permutation is unchanged, so reducing B's overlap count by one drops exactly the **last unit B had admitted** — its most marginal inclusion. Measured, the swap is one-for-one:

	review_id	meter	date	stratum
promoted to double review	**D0050**	ZN.10-PB.2	2025-05-31	C
released to single review	**D0043**	ZN.17-PB	2025-03-30	B

D0043 is still reviewed — by one reviewer instead of two. Nothing is dropped.

> **The C double review is a quality check, not an estimate.** With n = 1 there is no agreement to measure, and `agreement_report` refuses to produce one: any breakdown with fewer than `min_pairs_for_agreement` (5) pairs is returned as **NOT ESTIMABLE** with `kappa = NaN`, rather than printing a number that would read as 100% or 0% agreement. What the double review buys is the assurance that two people looked at the one item Model 3 surfaced and Model 1 cannot explain.

A double-reviewed unit keeps **one** `review_id` across both copies, so the two judgements join on it directly. Assignments live in the withheld key (`assigned_to`, `is_overlap`) and in `task4_reviewer_assignment.csv`.

17.3 Agreement reporting

Both statistics are reported, per sheet and per label column, because they answer different questions:

Statistic	What it says
raw agreement	the share of double-reviewed units the two reviewers labelled the same. Needs no assumption and is easy to read.
Cohen's kappa	agreement corrected for what chance would produce. Matters when one category dominates.
bootstrap interval on kappa	reported so the instability is visible rather than implied.

Both are **DIAGNOSTIC**. With 11 double-reviewed findings and 9 meter-days the intervals are very wide, and kappa is additionally subject to the *kappa paradox*: with one dominant category it can sit near zero at 90% raw agreement. They indicate whether the labelling task is reproducible enough to estimate from — not what the agreement is.

17.4 Identities — the remaining blocker

`reviewer_1`, `reviewer_2` and `adjudicator: TBD` are **placeholders**, and the issuance script treats them as a hard gate:

- `identity_status()` refuses to mark the package issuable;
- the workbooks are written with a `_DRAFT` **suffix**;
- the exit code is non-zero.

This is not bookkeeping. A placeholder reviewer id makes the 20 double-labelled units un-attributable, which is the one thing the overlap design exists to support; and an unnamed adjudicator leaves `Unsure` and disagreement with no owner.

Supplying the real names is a config edit plus one re-run of `issue_workbooks.py`. No unit is redrawn, no probability changes, and the assignment is reproduced identically from the seed.

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18. Status

ASSIGNMENT FROZEN — DRAFT pending named reviewer/adjudicator identities.

Component	State
200 sampled units, inclusion probabilities, allocation tables	frozen, verified unchanged
Reviewer assignment	complete, 20 overlap, seed-reproducible
Blinded workbooks	built, partitioned, label cells empty and unlocked
Un-blinded worklist	built, severity-ordered, all 405 findings
Withheld key	updated with assignments
Models 1, 2, 3	frozen, byte-identical
Reviewer and adjudicator names	**outstanding**
Precision, `unflagged_fault_rate`, Model 3 validation, agreement	**not computed** — no labels exist

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19. Assignment freeze, v6

The reviewer assignment is **FROZEN**. The adjustment changed who reviews two units and nothing else.

Sampled units	**200, unchanged** — none redrawn
Inclusion probabilities	**unchanged**, verified before and after
Allocation tables	**unchanged**
Models 1, 2, 3	**frozen, byte-identical**
Overlap	**20**: A 11, B 4, C 1, D 4
Workload	reviewer 1 **110**, reviewer 2 **110**
Reproducibility	identical on re-run from seed `20260926`; a different seed differs

A bug the adjustment surfaced

Naming the donor stratum as its own forced stratum made the forcing **silently cancel**: the +1 and the −1 offset exactly, C ended with no overlap, and nothing complained. The implementation now refuses that configuration with an explicit error, and a test locks it. It was never live in this package — the configured donor has always been B — but the failure mode would have been invisible.

Remaining blocker, unchanged

`reviewer_1`, `reviewer_2` and `adjudicator: TBD` are placeholders. The package stays `_DRAFT`, `issuable` is `false`, and the issuance script exits non-zero. Supplying the two real reviewer identifiers and the named adjudicator is a config edit plus one re-run; the assignment reproduces identically from the seed.

No precision, `unflagged_fault_rate`, Model 3 validation or agreement has been computed. All of them wait for returned labels.

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20. Precision-interval contract, v7

One documentation and test-contract cleanup. **Nothing was recomputed**: no unit redrawn, no assignment changed, no inclusion probability moved, no model rebuilt, and no statistical metric calculated — there are still no labels.

The v3 wording made the stratified bootstrap the primary interval and demoted linearisation to "an acceptable alternative". That was backwards, and it mattered: a with-replacement bootstrap across census cells invents sampling variance that does not exist.

Role	Method
Primary reported CI	survey-weighted linearisation with finite-population correction
Census cells (`pi_h = 1`)	contribute **exactly zero** sampling variance and are excluded from the variance sum
Secondary cross-check	FPC-aware stratified bootstrap: census cells held fixed, non-census cells rescaled by `sqrt(1 − f_h)`
Never reported	simple binomial `1.96*sqrt(p(1−p)/n)`
Unchanged	the weighted ratio point estimate, and the four-way `Unsure` bounds

Aligned in §7.3, §7.5, §8, the test table in §12, the Definition of Done, the decisions table, and the run metadata. The implementation in `src/review_loop/estimation.py` already matched this contract — `linearised_variance` is the primary and `fpc_stratified_bootstrap` the cross-check — so only the prose and the test wording were out of step.

Status: `ASSIGNMENT FROZEN — DRAFT pending named reviewer/adjudicator identities`